ESC113 TERM PROJECT

MANUAL & DOCUMENTATION

GROUP 9

2024-2025 SEMESTER II

TOPIC:

COMPUTATION OF LIQUID-PHASE COMPOSITION AND TEMPERATURE OF A VAPOUR-LIQUID MIXTURE

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Objective:

1. To compute Liquid-Phase Composition and Temperature of a Vapor-Liquid Mixture with given partial pressures.

2. To implement Newton-Raphson to get an estimate of volume-composition.

3. To compare between the convergence of different numerical methods namely, Newton's Method, Bisection Method and Regula Falsi for computation of temperature.

4. To implement a GUI-based interface i.e. an app with the MATLAB code and user-defined inputs.

Problem Statement:

A vapor-liquid mixture consists of two chemicals with known partial pressures at equilibrium.
The aim is to determine the composition of the liquid phase and the temperature of the mixture under the assumption of ideal gas and ideal liquid behaviour.
The saturation pressures of the components can be approximated using the Antoine equation:

$$log_{10}P_{sat} = A - \frac{B}{C + T}$$

where the saturation pressure P_{sat} is in (mmHg) and the temperature T is in Celsius.

The Antoine coefficients (A,B,C) for each component are provided.

The equilibrium condition follows Raoult's Law:

$$x_1 P_{sat,1} = y_1 P$$
$$x_2 P_{sat,2} = y_2 P$$

where:

- x_1, x_2 are the liquid-phase mole fractions,

- y_1, y_2 are the vapor-phase mole fractions,
- P is the total pressure of the system.

Methods to Solve:

Equations formed:

$$x_1 + x_2 = 1 \quad \rightarrow(1)$$

$$x_1 P_{sat,1} = y_1 P \quad \rightarrow(2)$$

$$x_2 P_{sat,2} = y_2 P \quad \rightarrow(3)$$

$$substituting\ y_1\ by\ \frac{P_1}{P}\ and\ y_2\ by\ \frac{P_2}{P},$$

$$P = x_1 P_1 + x_2 P_2$$

Therefore, the system of linear equations formed is:

$$f_1 = x_1 P_{sat,1} - y_1 P = 0$$
$$f_2 = (1 - x_1) P_{sat,2} - y_2 P = 0$$

Since the equations formed are multivariate systems of linear equations, they can only be solved using Newton-Raphson. Bisection Method and Regula Falsi apply for single-variable equations. Therefore, to provide a comparison between the methods, we will first get a reasonable estimate for x_1 using Newton-Raphson and then proceed to use that estimate for temperature calculation with Newton's Method, Bisection Method and Regula Falsi.

1. **Estimation of x_1 and T using Newton-Raphson Method :**

   Given a system of linear equations,

   R(X) = $[f_1(x_1, T), f_2(x_1, T)]^T$

Iterative Scheme: $J(X^{(k)}).(X^{(k+1)} - X^{(k)}) = -R(X^{(k)})$

Where,

- $X^{(k)}$ is the vector of unknowns at the k-th iteration.
- $J(X^{(k)})$ is the Jacobian matrix of partial derivatives of f_1 and f_2 with respect to x_1, x_2.
- $R(X^{(k)})$ is the residual function.

Convergence criterion: $\left|X^{(k+1)} - X^{(k)}\right| < tolerance$

2. Estimation of temperature along with comparison of different methods:

We will now reduce the equations to a single-variable equation in T:

$$f \quad = x_1 P_{sat,1} + (1 - x_1) P_{sat,2} - P$$

Where x_1 has been calculated using the Newton-Raphson method above. Therefore it is a single-variable equation in T. It can be solved using the following methods.

Methods for T:

a) Newton's Method [Fixed Point Method]

Given a function f(x)=0, it updates the solution iteratively.

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$$

Convergence criterion: $\left|x^{(k+1)} - x^{(k)}\right| < tolerance$

b) Bisection Method [Bracketing Method]

It checks the initial interval $[T_{low}, T_{high}]$ to see whether the root lies between them. $f(T_{low}).f(T_{high})<0$

$$s = \frac{T_{low} + T_{high}}{2}$$

If $f(T_{low}).f(s)$<0 : T_{high}= s
If $f(T_{high}).f(s)$<0 : T_{low}= s

Then proceed using the new interval $[T_{low}, T_{high}]$ and repeat the process.

Convergence criterion: $\left|\frac{T_{high}-T_{low}}{2}\right| < tolerance$

c) **Regula Falsi [Bracketing Method]**

It is similar to the bisection method. It checks the initial interval $[T_{low}, T_{high}]$ to see whether the root lies between them.
$f(T_{low}).f(T_{high})$<0
$$s = T_{high} - \frac{f(T_{high}).(T_{high}-T_{low})}{f(T_{high})-f(T_{low})}$$

If $f(T_{low}).f(s)$<0 : T_{high}= s
If $f(T_{high}).f(s)$<0 : T_{low}= s

Then proceed using the new interval $[T_{low}, T_{high}]$ and repeat the process.

Convergence criterion: $\left|x^{(k+1)} - x^{(k)}\right| < \text{tolerance}$

Dew-Point Bubble Point Calculations:

Bubble Point:
It is the temperature (at a given pressure) where the first bubble of vapor is formed when heating a liquid consisting of two or more components.

Using the equation
$$P = x_1 P_{sat,1} + (1 - x_1) P_{sat,2}$$
Taking x_1 as an array of numbers from 0 to 1 for linearly spaced mole fractions, calculation for T using Newton's Method. The initial guess of T is the same as before.
The process involves:

1. Guessing an initial temperature T0.
2. Calculating the saturation pressures for the guess temperature using the Antoine equation.
3. Computing the total pressure from the mole fraction and the saturation pressures.
4. Updating the temperature guess until the calculated total pressure P matches the specified pressure.

Dew Point:

The dew point is the temperature and pressure at which the first drop of vapour condenses into a liquid.

Using the equation,

$$1 = \frac{y_1 \mathrm{P}}{P_{sat,1}} + \frac{(1 - y_1)\mathrm{P}}{P_{sat,2}}$$

Taking y_1 as an array of numbers from 0 to 1 for linearly spaced mole fractions, calculation of T using Newton’s method. The initial guess for T is same as before.

1. Guess an initial temperature T0.
2. Calculate the saturation pressures at the guessed temperature using the Antoine equation.
3. Calculate the vapor-phase mole fractions for both components.
4. Update the temperature guess using Newton’s Method until the left-hand side of the equation matches 1.

APP OVERVIEW:

Aim:

- A MATLAB-based App that provides an interactive and user-friendly platform for solving the VLE
- To depict 3 classical root-finding methods Newton-Raphson, Bisection, and Regula Falsi
- To provide a comparison of convergence methods and accuracy using plots
- To understand the performance difference by changing initial conditions and tolerance

User Defined Inputs:

- **Choice of Chemicals:**

  We have implemented two dropdowns with a choice of 4 chemicals in total. These include: Acetone, Ethylbenzene, Toluene and Benzene. If the user picks the same chemicals twice, the app asks the user to pick different values.

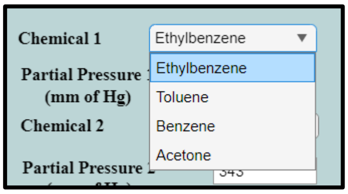


Figure 1: Dropdown for choice of chemical

- **Partial Pressures:**

  The user can input the initial partial partial pressures for each of the two chemicals.

- **Tolerance Value:**

  The user can input the tolerance value of the calculation of convergence

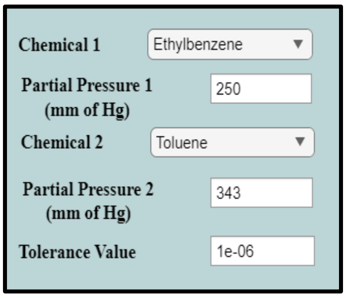


Figure 2: Options to input pressure and dropdowns for chemicals

- **Temperature:**

  The user can change the initial guess temperature for each of the methods. The Regula Falsi and the Bisection Method add 60 or subtract 40 to get a reasonable estimate.

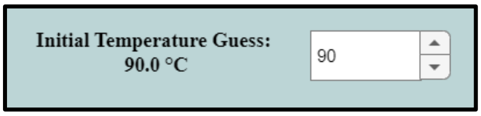


Figure 3: Initial Temperature Guess Spinner

- **Type of Method:**

  The user has an option to choose between the type of error convergence plot they want to see. The four options are: Newton's Method, Bisection Method, Regula Falsi and the original Newton Raphson Method.

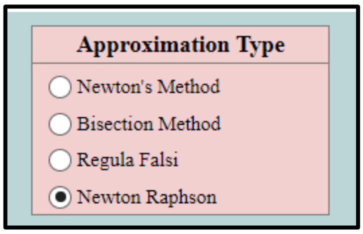


Figure 4: Showing the Radio button group for Approximation type

- **Plot Button**
  The plot button allows the calculation to take place and the user to see the visualization and the results in the respective area

- **Clear Plot Button:**
  The user can create as many error convergence plots they want for each of the methods on the same graph for comparison. Using this button, they can clear the axes and start again.

Figure 5

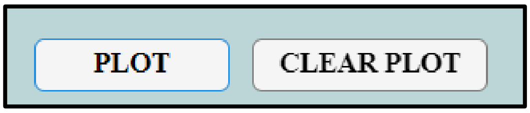


Figure 5: Showing PLOT and CLEAR PLOT buttons

Figure 6: Empty plot after pushing CLEAR PLOT button

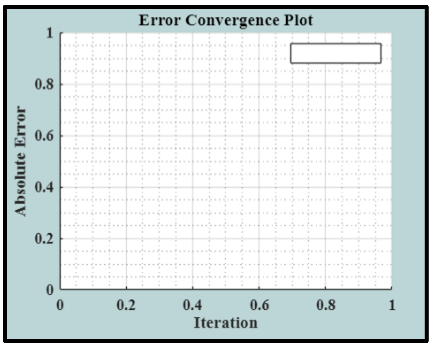


Figure 6

- **Dew Point- Bubble Point Curve button:**
  Using this button, the user can plot the dew-point curve and the bubble point curve at the given total pressure and at the calculated temperature

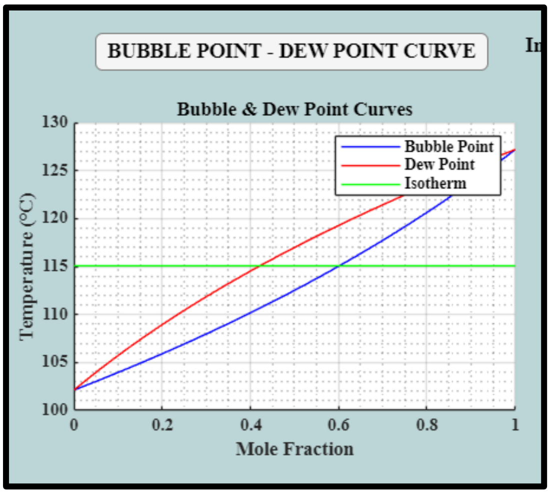


Figure 7

Figure 7: This shows the button and a sample dew point-bubble point curve at the calculated temperature along with the isotherm of the calculated temperature

FEATURES:

1. Comparison Plotting:

The user can go on selecting different methods and changing the guess temperatures. The convergence plots for each will be made on the same graph for comparison and better visualization. The user can use CLEAR PLOT button to clear the axes.

Convergence plots have been plotted to visualise the rate of error reduction over iterations for each method.

- X-axis: Iteration number
- Y-axis: Error magnitude

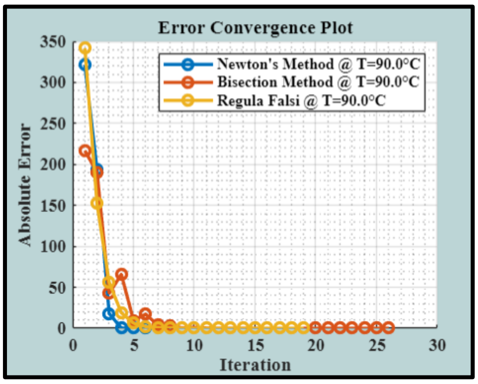


Figure 8: Error Convergence comparison plots

2. Output Display:

Once the user changes the values and pushes the plot button, the results are outputs information about the calculation at that guess temperature using all the methods.

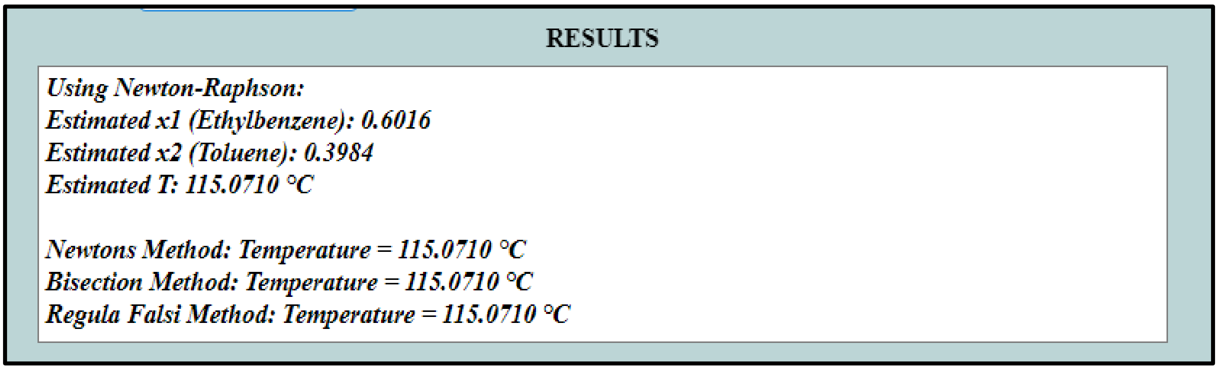


Figure 9: Snapshot of the Result area after inputting default values

3. Result Area:

a) Estimated liquid phase composition

The estimated mole fractions in the liquid phase of a binary mixture of the selected chemicals

- x1
- x2=1-x1

These are **calculated using Newton-Raphson method** by matching the partial pressures with Raoult's Law. We use Antoine's equations to calculate P_sat at each iteration using the temperature guess.

b) Estimated Temperature Results using the estimated x1:

- Temperature estimated from Newton's method
- Temperature estimated from Bisection method
- Temperature estimated from Regula Falsi method

4. The Dew point-Bubble point curve

The user can choose to plot the dew point-bubble point curve for the given configurations.

CONSTRAINTS:

1. If 2 same chemicals are chosen then the problem is redundant. Therefore, the program displays an appropriate warning

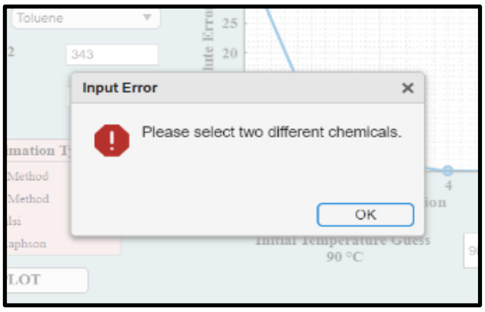


Figure 10: Warning when two same chemicals are input.

2. If the value of pressure input is negative the program shows a warning and resets to the default configuration.

Figure 11: Showing warning when the pressure inputted is out of range

3. The tolerance value has a range of 1E-12 to 0.1. If the user inputs a value out of this range, then the program resets the value to the default configuration i.e. 1E-6

4. The temperature guess spinner has a range between 30 and 170 degrees Celsius. If the user inputs a value outside this range, the programs resets it to 90.

REFERENCE SAMPLE PROBLEM: Default Configuration

*The reference problem used is: Problem 3.26, Page Number 197, Dorfman, Kevin D. and Daoutidis, Prodromos, Numerical Methods with Chemical Engineering Applications, Cambridge University Press: New Delhi, 2017.

3.26 A vapor–liquid mixture of ethylbenzene and toluene has a partial pressure of 250 mm Hg of ethylbenzene and 343 mm Hg of toluene. Write a MATLAB program that will compute the composition of the liquid phase and the temperature of this mixture if we assume ideal gas and liquid behavior. You can approximate the saturation pressures using the Antoine equation

$$\log_{10} P^{\text{sat}} = A - \frac{B}{T + C}$$

where the saturation pressure is in mm Hg and the temperature is in Celsius. The Antoine coefficients are in the table below.

Chemical	A	B	C
Ethylbenzene	6.957 19	1424.255	213.21
Toluene	6.954 64	1344.8	219.48

Your program should output the temperature and liquid-phase mole fractions of each species. What are the equations used for the residual and the Jacobian?

Figure 12: Screenshot from Dorfman of the reference problem

According to this, the configurations are as follows:

- Chemical 1 - Ethylbenzene
- Partial Pressure 1(mm of Hg) - 250
- Chemical 2 - Toluene
- Partial Pressure 2(mm of Hg) - 343
- Tolerance Value – 1E-06

RESULTS

Using Newton-Raphson:
Estimated x1 (Ethylbenzene): 0.6016
Estimated x2 (Toluene): 0.3984
Estimated T: 115.0710 °C

Newtons Method: Temperature = 115.0710 °C
Bisection Method: Temperature = 115.0710 °C
Regula Falsi Method: Temperature = 115.0710 °C

Figure 13: Result output of the sample problem

Error Convergence Plots for the Sample Problem:

1. **Regula Falsi:**

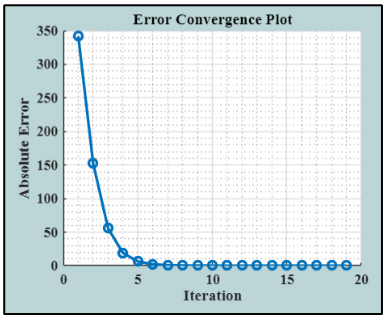


Figure 14: Error Convergence plot for Regula Falsi

2. **Newton's Method:**

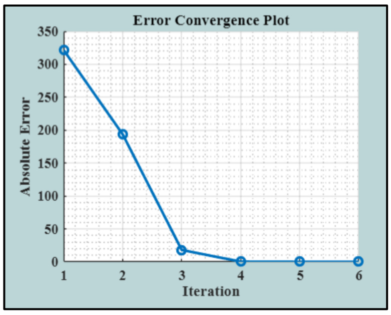


Figure 15: Error Convergence plot for Newton's Method

3. **Bisection Method:**

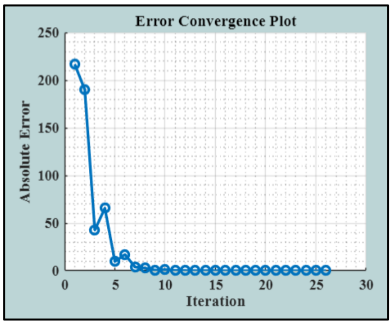


Figure 16: Error Convergence plot for Bisection Method

4. **Newton Raphson Error for X0:**

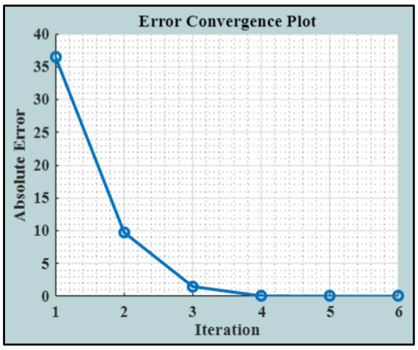
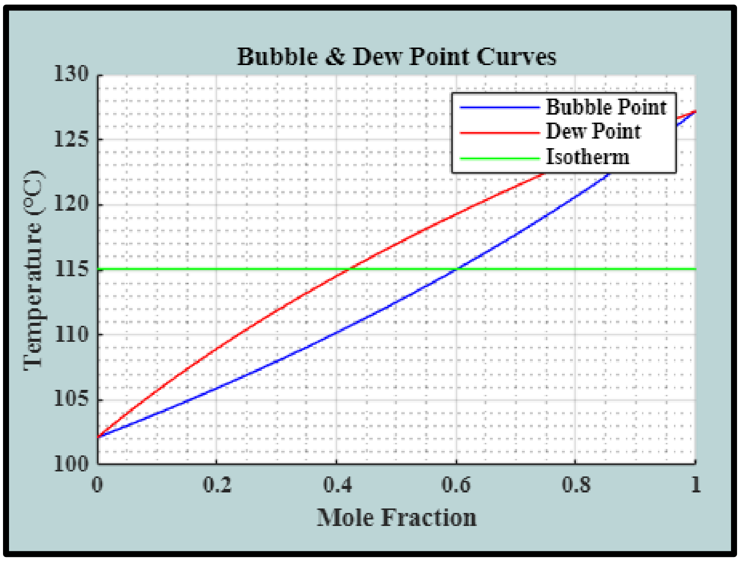


Figure 17: Error Convergence plot for Newton Raphson of the combined X vector $\begin{bmatrix} x1 \\ T \end{bmatrix}$

Bubble Point-Dew Point Curve for the default configuration



This plot shows the bubble point a curve and the dew point curve at total pressure P1+P2 and the isotherm of the calculated mixture temperature.

COMPARITIVE ANALYSIS OF THE METHODS

Method	Strengths	Weaknesses
Newton-Raphson (Multivariate)	- Fast convergence near the root - Handles systems of nonlinear equations	- Requires good initial guess - Needs computation of Jacobian matrix
Newton's Method (Single-variable)	- Fast convergence when derivative is well-behaved	- May diverge if derivative is small or initial guess is poor
Bisection Method	- Guaranteed convergence - Simple and robust	- Slower convergence - Requires bracketing the root
Regula Falsi	- Typically faster than Bisection - Does not need derivative	- Can stagnate if one end of interval does not move

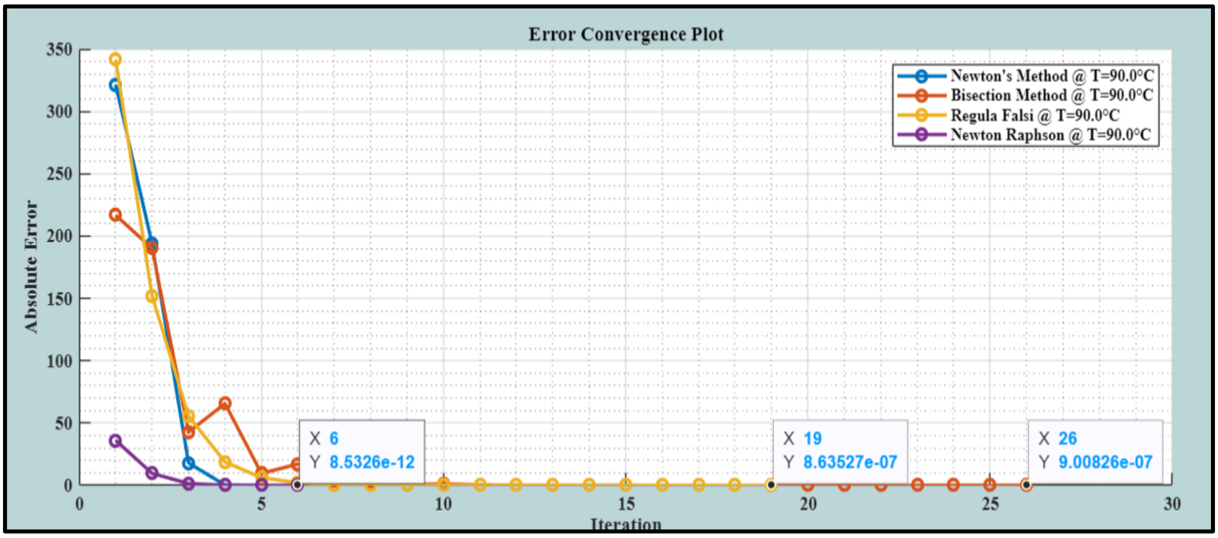


Figure 13: Error Convergence plot for comparison

Observations from Comparative Graph:

1. **Convergence Speed:**
   - Newton's Method and Newton-Raphson converge very quickly, reaching very low errors ($\sim 10^{-12}$) within 6 iterations.
   - Bisection and Regula Falsi take much longer (~19 to 26 iterations) and only reach error magnitudes around $\sim 10^{-7}$, showing slower convergence.

2. **Accuracy:**
   - Newton's Method gives the highest precision, reaching an absolute error of $\sim 8.5 \times 10^{-12}$.
   - The Bisection and Regula Falsi methods plateau at a higher error level ($\sim 10^{-7}$)), meaning they are less accurate.
   -

3. **Stability:**
   - Bisection and Regula Falsi show some oscillations or inconsistent reductions in error in early iterations.
   - Newton-based methods show smooth and rapid decline in error, indicating better numerical stability for this problem.

4. **Efficiency:**
    - Fewer iterations needed for Newton's methods imply lower computational cost when derivative calculations are not expensive or unstable.
    - Bisection/Regula Falsi methods, while more robust, are inefficient in convergence speed.

CONCLUSION

- **Newton's Method and Newton-Raphson** are **superior** in both speed and accuracy for this case at 90°C.

- **Bisection and Regula Falsi** are **slower and less precise**, but may still be preferable when derivatives are hard to compute or the initial guess is poor.

- The **error tolerance** $\sim 10^{-6}$, goal was met by all methods, but with **extremely different iteration counts**.

FUTURE USE OF CODE AND APPLICATIONS IN CHEMICAL ENGINEERING:

1. Demonstration of Numerical Methods:
   The code can be used to compare and visualize convergence behaviour and performance of different root finding methods

2. Extension to support bubble-point and dew-point temperature calculations and building equilibrium stage models such as multicomponent VLE other than just visualization.

3. Can be used to estimate thermodynamic behaviour from Antoine constants and partial pressures by extension to take user-defined Antoine constants or import from the NIST database.
4. Can be upgraded to support non-ideal systems and activity coefficient models.

POTENTIAL IMPROVEMENTS:

1. Addition of more chemicals or input of Antoine coefficients
2. Integration of activity coefficient models
3. Extension to a multicomponent system
4. Extension to support non-ideal systems

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